

Strapi v5 Rollback Checklist

Odyssey · ODY-596 · ASD-STE100 Simplified Technical English version of `docs/playbooks/strapi-v5-rollback.md`
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About this checklist

Use this checklist if the cutover from Strapi v4 to Strapi v5 fails. The checklist puts Strapi v4 back.

Do the steps in sequence. Put a mark in each box when you complete the step.

The checklist applies to the production environment and to the dev environment. For dev, use the dev names. For dev, use the `develop` branch, not the `production` branch.

The reasons for each step are in the Linear ticket [ODY-596](#). The steps for the cutover are in `docs/playbooks/strapi-prod-migration-day.md`.

Terms

Term	Description
Backend	The Strapi server. It reads and writes the database.
Frontend	The Next.js website that users see. It gets data from the backend.
ECS	Amazon Elastic Container Service. It runs the backend and the frontend.
Task	One copy of the backend or of the frontend that runs in ECS.
Stop the backend	Set the number of backend tasks to 0. When no task runs, no task can change the database.
Task definition	The instructions that ECS uses to start a task. The instructions include the version of the software.
Rollback task definition	A task definition that starts the Strapi v4 version. The Driver makes it at the start of the window.
ARN	Amazon Resource Name. It is the full name of an item in AWS.
RDS	Amazon Relational Database Service. It operates the database server.
Migration	The changes that Strapi v5 makes to the database when it starts for the first time. You cannot undo these changes.
Frozen copy	A copy of the database that the Driver makes before the migration. Nothing changes the frozen copy.
Fingerprint	A report that shows the number of rows in each database table. The <code>flavor</code> line in the report shows if the database is <code>v4</code> , <code>v5</code> or <code>PARTIAL</code> .
Snapshot	A full backup of the database server in RDS.
Dump	A backup file of the database. It is on the encrypted computer of the Driver.
Smoke test	A short set of checks of the website. Refer to the section "Smoke test".
Window	The maintenance period. Users cannot use the website during the window.
T0	The start of the window.
T_open	The time when the website opened to users again after the cutover. The time is in UTC (Coordinated Universal Time).
Window notes	The values that the Driver writes at the start of the window.

Term	Description
Pipeline	The GitHub Actions workflow. It builds and deploys the software when a person pushes to the <code>production</code> branch.
Driver	The person who does the steps.
Approver	The person who must approve a rollback after <code>T_open</code> . A rollback after <code>T_open</code> deletes the work of users. Thus, a second person must approve it.
Comms	The person who sends messages to users and to the team.

Have these values before you start

Make sure that the window notes contain these values:

- ☐ The names of the ECS cluster, the backend service and the frontend service
- ☐ The usual number of frontend tasks
- ☐ The database name `<db>` and the name of the frozen copy `<db>_v4_frozen`
- ☐ The ARN of the backend rollback task definition
- ☐ The ARN of the frontend rollback task definition
- ☐ The ID of the T0 snapshot
- ☐ The path of the dump file and its SHA-256 value
- ☐ The T0 fingerprint file
- ☐ The fingerprint file of the frozen copy
- ☐ `T_open`. If the website did not open again, this value is empty.
- ☐ The name of the media bucket and the list of media files at T0
- ☐ The names of the Driver and the Approver

If a value is not in the window notes, find it before you continue.

The `psql` commands use the `PGHOST`, `PGUSER` and `PGPASSWORD` values. Set these values for the RDS database server. Use the PostgreSQL 18 client tools.

Make sure that the scripts in `scripts/ody-596/` are in the repository. If the scripts are not there, you cannot use this checklist.

Cautions for all procedures

CAUTION: DO NOT START STRAPI V4 ON A DATABASE THAT STRAPI V5 CHANGED. Strapi v4 deletes all links between records, and it does not show an error. Examples of links: the lessons in a droplet, the droplets in a voyage, enrollments and group members. First, put back the frozen copy or a backup.

CAUTION: PUT BACK THE FRONTEND AND THE BACKEND TOGETHER. The v5 frontend cannot save data to Strapi v4. The v4 frontend cannot save data to Strapi v5.

CAUTION: ALWAYS USE THE ROLLBACK TASK DEFINITION TO START THE BACKEND. Do not only set the number of tasks to 1. The pipeline uses only the `:latest` image, and this image is now Strapi v5.

CAUTION: MONITOR EACH V4 DEPLOY UNTIL ECS SHOWS THAT IT IS HEALTHY. If the v4 backend tasks fail again and again, stop the backend immediately. If you do not, ECS starts Strapi v5 again. Strapi v5 then changes the database that you put back.

CAUTION: DO NOT DO A `terraform apply` UNTIL 24 HOURS AFTER T_OPEN. A `terraform apply` changes the number of tasks.

CAUTION: DO NOT USE THE `docker compose up` COMMAND ON YOUR COMPUTER. The local Docker stack can connect to the production database.

CAUTION: DO NOT DELETE THE FAILED DATABASE, THE FROZEN COPY, THE SNAPSHOT OR THE DUMP. Get approval from the team first.

1. Decide if a rollback is necessary

Before T_open

The Driver decides. A rollback before T_open does not delete the work of users.

Do the rollback if one of these conditions occurs:

- ☐ Strapi v5 does not complete its first start in the time limit in the window notes.
- ☐ The log shows a migration error.
- ☐ After the migration, the row counts in the fingerprint are different from the rehearsal.
- ☐ The v5 `03` check fails.
- ☐ A smoke script shows `FAIL`.
- ☐ A smoke test item fails, and you cannot repair it in 15 minutes.
- ☐ The hard deadline in the window notes occurs, and the smoke test did not pass. For this condition, always do the rollback.

If the migration fails, do not try the migration again in the same window.

After T_open, for a maximum of 24 hours

The Approver must approve the rollback. A rollback after T_open deletes all the work that users saved after T_open.

Do the rollback if one of these conditions occurs:

- ☐ Data is incorrect or is not there. Examples: lessons are not in their droplets, or enrollments are not there. For this condition, always do the rollback.
- ☐ Users with one role cannot sign in for 30 minutes, and you cannot find a repair.
- ☐ Many users cannot save enrollments or progress.
- ☐ Staff cannot edit content or upload media, and you cannot repair it before the end of the work day.
- ☐ The backend shows 5xx errors at more than 2 times the usual rate for 30 minutes.

If the data is correct and a code repair is possible in about 2 hours, repair the code. Do not do a rollback.

More than 24 hours after T_open

Do not do a rollback. Repair the fault in the code.

Record the decision

- ☐ Write the time in UTC, the condition that caused the rollback, and the name of the person who approved it.

2. Select the procedure

Condition	Procedure
Strapi v5 did not start yet.	A. Stop the cutover
Strapi v5 started 1 time or more.	B. Put back the frozen copy. This is the usual procedure.
You must do procedure B, but the frozen copy is not there or its check fails.	C. Put back the dump
The database server does not operate.	D. Put back the snapshot on a new server

If you do not know if Strapi v5 started, get a fingerprint of the database:

```
psql -d <db> -f scripts/ody-596/fingerprint.sql
```

- If the `flavor` value is `v4`, do procedure A.
- If the `flavor` value is `v5` or `PARTIAL`, do procedure B.

Procedure A: Stop the cutover

Use this procedure if Strapi v5 did not change the database.

- ☐ If the pipeline deployed the new frontend, deploy the frontend rollback task definition:

```
aws ecs update-service --cluster <cluster> --service <frontend-service> --task-definition <frontend rollback ARN>
```
- ☐ Deploy the backend rollback task definition with 1 task:

```
aws ecs update-service --cluster <cluster> --service <backend-service> --task-definition <backend rollback ARN> --desired-count 1
```
- ☐ Monitor the backend and the frontend until ECS shows that they are healthy:

```
aws ecs wait services-stable --cluster <cluster> --services <backend-service> <frontend-service>
```
- ☐ Do the [smoke test](#).
- ☐ Do the steps in [After each rollback](#). The data did not change. Thus, do not do steps 4 and 5 in that section.

Procedure B: Put back the frozen copy

The time for this procedure is in the window notes. The swap of the 2 databases takes some seconds.

- ☐ Stop the backend:

```
aws ecs update-service --cluster <cluster> --service <backend-service> --desired-count 0
```
- ☐ If T_{open} occurred, stop the frontend. This stops new changes from users.

```
aws ecs update-service --cluster <cluster> --service <frontend-service> --desired-count 0
```
- ☐ Wait until no tasks run.
- ☐ Connect to the `postgres` database. Close the other connections to `<db>` and to `<db>_v4_frozen`:

```
SELECT pg_terminate_backend(pid) FROM pg_stat_activity WHERE datname IN ('<db>', '<db>_v4_frozen') AND pid <> pg_backend_pid();
```
- ☐ If T_{open} occurred, save the list of changes after T_{open}. You use this list later to enter the work again.

```
psql -d <db> -v t_open='<T_open UTC>' -f scripts/ody-596/writes-since.sql > ~/ody-596/<env>-writes-since-<date>.txt
```
- ☐ Get a fingerprint of the frozen copy:

```
psql -d <db>_v4_frozen -f scripts/ody-596/fingerprint.sql
```
- ☐ Make sure that the `flavor` value is `v4`. Make sure that the row counts are the same as in the fingerprint file of the frozen copy. If one of these checks fails, stop and do procedure C.
- ☐ Swap the databases. The script changes the name of `<db>` to `<db>_v5_failed`. Then it changes the name of `<db>_v4_frozen` to `<db>`.

```
psql -d postgres -v live=<db> -v frozen=<db>_v4_frozen -v failed=<db>_v5_failed -f scripts/ody-596/swap-back.sql
```

9. ☐ Get a fingerprint of `<db>`. Make sure that the `flavor` value is `v4`.

```
psql -d <db> -f scripts/ody-596/fingerprint.sql
```
10. ☐ Make sure that the owner and the settings are the same as in the T0 fingerprint. The swap script shows these values.
11. ☐ Deploy the backend rollback task definition with 1 task:

```
aws ecs update-service --cluster <cluster> --service <backend-service> --task-definition <backend rollback ARN> --desired-count 1
```
12. ☐ Deploy the frontend rollback task definition with the usual number of tasks:

```
aws ecs update-service --cluster <cluster> --service <frontend-service> --task-definition <frontend rollback ARN> --desired-count <usual number>
```
13. ☐ Monitor the backend and the frontend until ECS shows that they are healthy. If the backend tasks fail again and again, stop the backend immediately.
14. ☐ Do the [smoke test](#).
15. ☐ Do the steps in [After each rollback](#).

Procedure C: Put back the dump

Use this procedure if the frozen copy is not there or is damaged. The time for this procedure is in the window notes.

This procedure puts back the data from the start of the window. This data is from before the ODY-633 changes. Strapi v4 can use this data.

1. ☐ Do steps 1 to 5 of procedure B.
2. ☐ Get the SHA-256 value of the dump file. Make sure that it is the same as the value in the window notes.

```
shasum -a 256 <dump file>
```
3. ☐ Make an empty database:

```
createdb <db>_v4_restored
```
4. ☐ Put the dump into the new database:

```
pg_restore --no-owner --role=<db owner> -j 4 -d <db>_v4_restored <dump file>
```
5. ☐ Get a fingerprint of `<db>_v4_restored`. Make sure that the `flavor` value is `v4`.

```
psql -d <db>_v4_restored -f scripts/ody-596/fingerprint.sql
```
6. ☐ Make sure that the row counts are the same as in the T0 fingerprint file.
7. ☐ Swap the databases:

```
psql -d postgres -v live=<db> -v frozen=<db>_v4_restored -v failed=<db>_v5_failed -f scripts/ody-596/swap-back.sql
```
8. ☐ Continue at step 9 of procedure B.

Procedure D: Put back the snapshot on a new server

Use this procedure only if the database server does not operate. This is the slowest procedure. It can take 30 minutes or more.

1. ☐ If you can connect to the database, do steps 1 to 5 of procedure B.
2. ☐ If you cannot connect to the database, stop the backend and the frontend. Do not do the other steps of procedure B.
3. ☐ Make a new database server from the T0 snapshot. Use the same subnet group, security groups and parameter group as the old server.

```
aws rds restore-db-instance-from-db-snapshot --db-instance-identifier <new-db-instance> --db-snapshot-identifier <T0 snapshot id> --db-subnet-group-name <subnet group> --vpc-security-group-ids <sg> --db-parameter-group-name <parameter group>
```
4. ☐ Wait until the status of the new server is `available`.
5. ☐ Get a fingerprint of the database on the new server. Make sure that the `flavor` value is `v4`.
6. ☐ Write down the `DATABASE_HOST` value. This value is in the backend secret in AWS Secrets Manager.
7. ☐ Change the `DATABASE_HOST` value to the address of the new server.
8. ☐ Continue at step 11 of procedure B.
9. ☐ Tell the team that the infrastructure is now different from the Terraform files. Do not do a `terraform apply` until the team corrects this difference.

Smoke test

Do these checks as a staff user. Then do the checks again as a user who is not staff.

- ☐ Sign in with each role.
- ☐ Open the Explore page, a droplet page, a lesson page and a voyage page.
- ☐ Make sure that each droplet shows its lessons. Make sure that each voyage shows its droplets.
- ☐ Enroll in a droplet or a voyage.
- ☐ Complete a part of a lesson.
- ☐ Edit a droplet on the website.
- ☐ Edit an entry in the Strapi admin panel.
- ☐ Upload an image.
- ☐ Open a group page that has due dates.
- ☐ Open the admin dashboard.

After each rollback

CAUTION: DO NOT PUSH TO THE `production` **BRANCH UNTIL STEP 2 IS COMPLETE.** A push starts the pipeline, and the pipeline does the Strapi v5 migration again. For dev, this caution applies to the `develop` branch.

1. ☐ Tell the team to stop all pushes to the `production` branch. For dev, use the `develop` branch.
2. ☐ Revert the v5 merge on the `production` branch, and push the revert. A person with push access does this step.

```
git revert -m 1 <v5 merge commit>
```

The pipeline then builds the Strapi v4 images again. This is safe because the database is v4 again.
3. ☐ Comms tells users that the website uses the old version again. If `T_open` occurred, Comms tells users that the changes from `<T_open>` to `<rollback time>` are lost.
4. ☐ Use the list of changes after `T_open`. Enter the important work again by hand: staff content edits, new droplets, new lessons, media and group changes. Do not enter enrollments or progress again. The message in step 3 tells users about them.
5. ☐ Compare the media bucket with the list of media files at `T0`:

```
aws s3 ls s3://<media-bucket> --recursive > ~/ody-596/<env>-s3-after-rollback.txt
```

Keep the files that users added during Strapi v5. These files cause no problems. If a person deleted a file during Strapi v5, the pages that use the file are broken. Put back the file from S3 versioning.
6. ☐ If users who are signed in see errors, do the session procedure from ODY-596, question Q4. The team did not write this procedure yet. The rehearsal will do a test of it.
7. ☐ Keep `<db>_v5_failed`, the snapshot and the dump until the team finds the cause. Get approval before you delete one of them.
8. ☐ Write a report in ODY-603. Include the times, the condition that caused the rollback, the duration of each step and the faults. Do not include real user data.
9. ☐ Make a ticket for the cause of the failure.
10. ☐ Before you try the cutover again, do the full cutover procedure from the start. Make a new frozen copy.
11. ☐ After the next migration, start the frontend again. Strapi v5 gives each record a new ID at each migration.